Deep Learning Based Driver Drowsiness & Vigilance Detecting System

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Introduction

VISUAL BEHAVIOUR

Signs of reduced vigilance can appear through eye closure, yawning, head movement and facial changes.

DRIVER DROWSINESS

Drowsiness reduces attention and reaction capability, creating a major road-safety concern.

OUR APPROACH

Use video and deep learning to learn relevant visual patterns automatically.

Problem statement

The objective is to develop and evaluate an automated system capable of interpreting a driver's facial behaviour from video and determining whether the driver is Alert, at Low Vigilance or Drowsy. The system should use a camera/video stream as its primary input and should not require physical sensors attached to the driver.

OBJECTIVE

1. Develop a non-contact video-based driver-state monitoring system.
2. Automatically learn relevant facial and behavioral patterns from video frames.
3. Classify the driver as Alert, Low Vigilance or Drowsy.
4. Capture temporal behavior rather than relying only on single-frame decisions.
5. Reduce dependence on manually engineered features such as EAR-only rules.
6. Evaluate generalization on completely unseen participants.
7. Create a foundation that can later be converted into a real-time monitoring system.

Data Set

- **UTA Real-Life Drowsiness Dataset (UTA-RLDD), University of Texas at Arlington.**

60	180	~30 h	3
Participants	Videos	RGB Video	Classes

ALERT

Normal attentive driving state.

LOW VIGILANCE

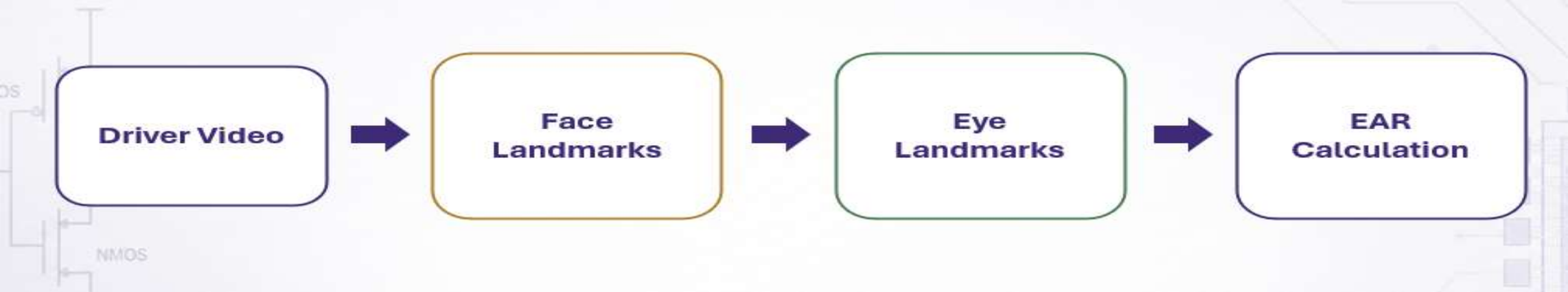
Reduced alertness / transition state.

DROWSY

Clear signs of reduced vigilance.

Dataset is organized by participant and vigilance state.

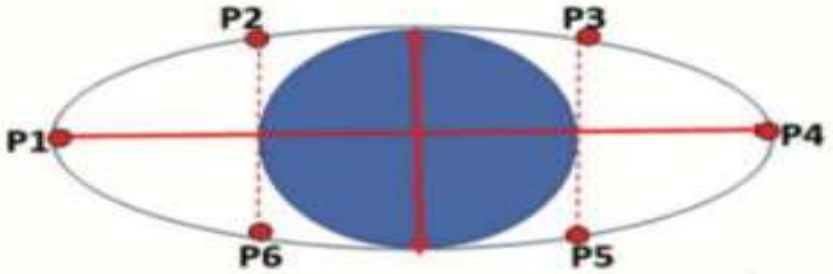
Initial research direction

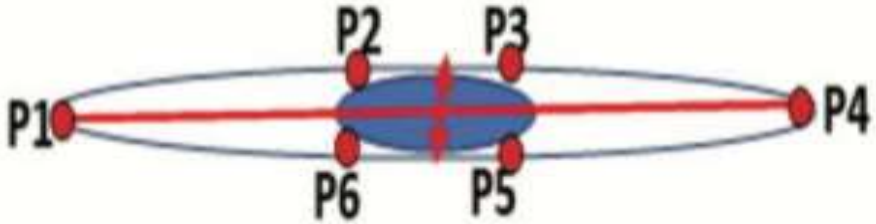


Formula Used

$$EAR = ||P2-P6|| + ||P3-P5|| \ / \ 2 * ||P1-P4||$$

$$EAVG\ EAR = 1 / 2\ (EAR\ Left + EAR\ Right)$$









EYE LANDMARKS

Six landmark points are selected for each eye.

EAR

Vertical eye distances are compared with horizontal eye width to obtain a normalized measure.

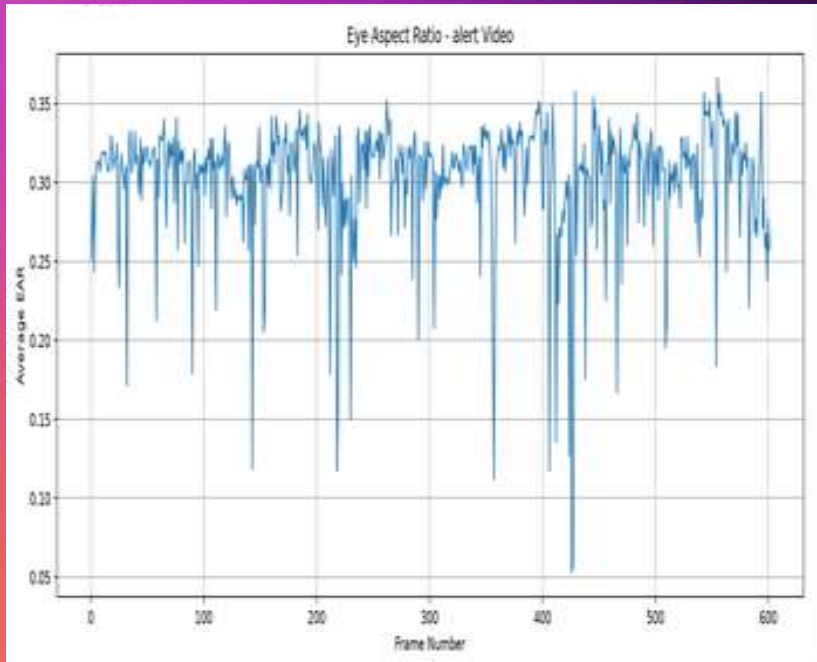
FRAME-BY-FRAME

EAR is calculated repeatedly across video frames to observe changes over time.

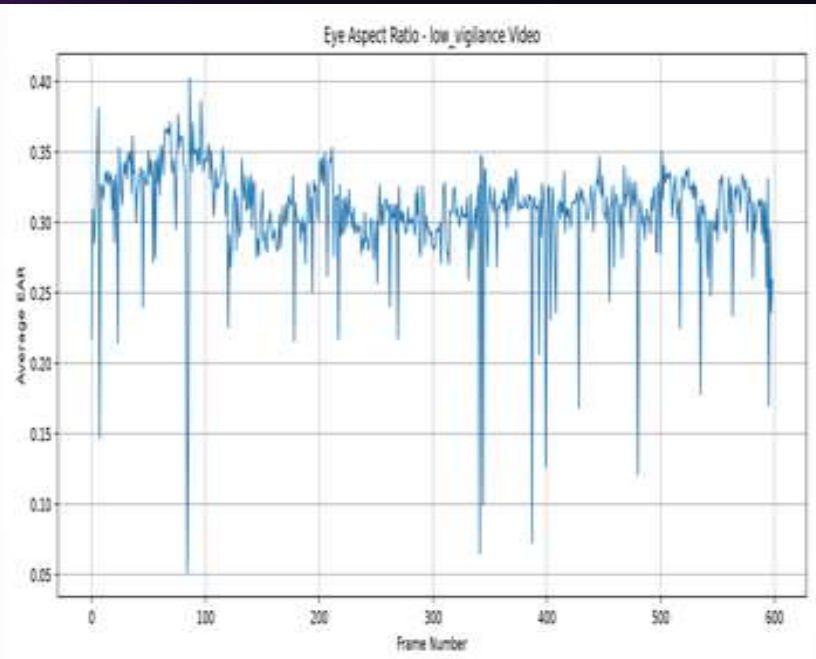
Work done And EAR Calculation

Observed EAR Behaviour

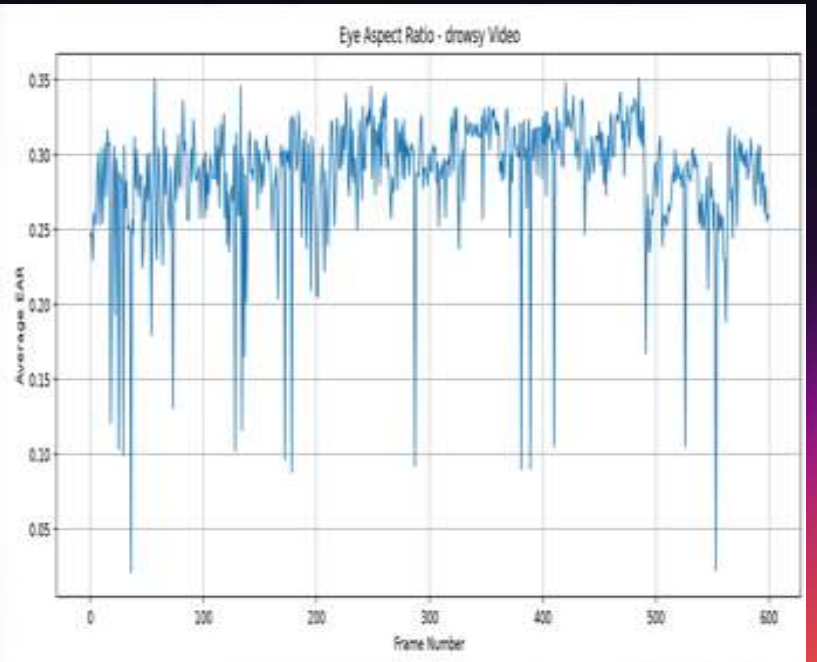
EAR values plotted across video frames



ALERT



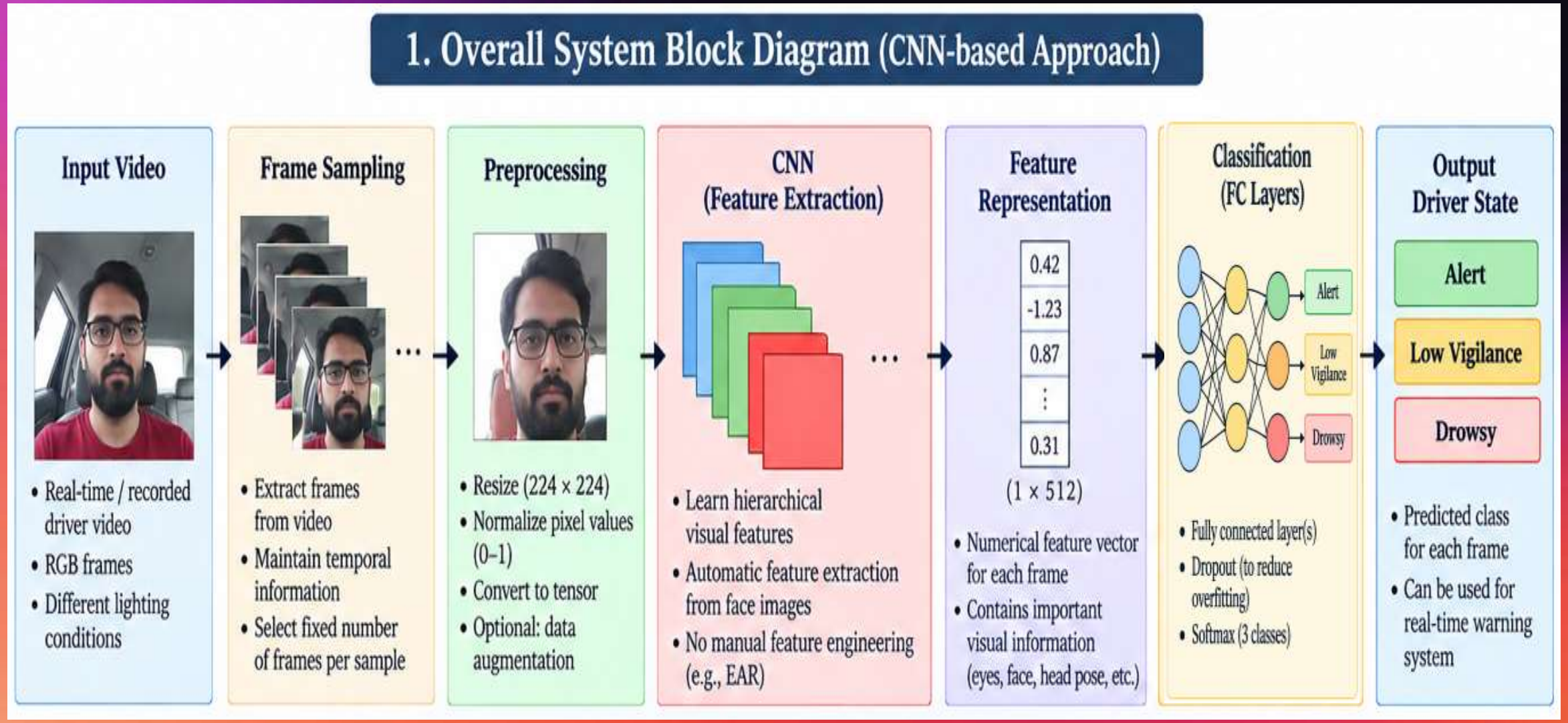
LOW VIGILANCE



DROWSY

The plots show that eye behaviour changes over time, but the signal is not perfectly clean or sufficient by itself to describe the complete driver state.

Proposed Model



Application

1. Driver Safety & Accident

Prevention Detects signs of reduced vigilance and can warn the driver before the situation becomes dangerous.

2. Real-Time Driver Monitoring

Can be integrated with an in-car camera to continuously monitor the driver's visual behaviour.

3. Fleet & Commercial Vehicles

Useful for monitoring drivers of buses, trucks, taxis and delivery vehicles, where long driving hours can increase fatigue.

4. Driver Alert / Warning System

The predicted state can trigger an alarm, buzzer or visual warning when drowsiness is detected.

5. Advanced Driver Assistance Systems (ADAS)

Can act as one component of a larger vehicle safety system alongside other driver-monitoring functions.

6. Long-Distance & Night Driving

Particularly useful during situations where drivers are more likely to experience fatigue, such as long highway journeys or night driving.

Reference

- Soukupová, T., & Čech, J. (2016). Real-Time Eye Blink Detection using Facial Landmarks.
- Ghoddousian, R., Rosenberg, M., & others. A Robust Deep Learning-Based System for Driver Drowsiness Detection.
- UTA Real-Life Drowsiness Dataset (UTA-RLDD), University of Texas at Arlington.
- OpenCV — Computer Vision Library.
- PyTorch — Deep Learning Framework.